



COLUMBIA UNIVERSITY

Department of Statistics

Completed in Partial Fulfillment of the Requirements of

STAT S4204

Statistical Inference

Summer 2026

ASSIGNMENT 9

Daniel Rabinowitz

dr105@columbia.edu

Tyler Sotomayor

tyler.sotomayor@columbia.edu

Contents

HPS, Chapter 2: Exercises 1–61	1
Exercise 1: An Unbiased Estimator of the Normal Variance (Known Mean Zero) .	1
Exercise 2: Scaling a Single Uniform Observation to Unbiasedness	2
Exercise 3: Unbiased Estimation for a Gamma Shape	3
Exercise 4: When a Linear Combination Is Unbiased for the Mean	4
Exercise 5: The Bias of the Minimax Binomial Estimator	5
Exercise 6: The Minimax Estimator Has Constant Risk	6
Exercise 7: The Negative Binomial and an Unbiased Estimator of p	7
Exercise 8: The Sample Mean Is Efficient for the Poisson	9
Exercise 9: Best Weights for a Linear Unbiased Estimator	10
Exercise 10: The Efficiency of Using Only Two Observations	12
Exercise 11: The Mean Minimizes Expected Squared Distance	13
Exercise 12: The Variance Bound for the Cauchy Location	14
Exercise 13: Why the Bound Fails for the Uniform	15
Exercise 14: Mean and Variance of the Known-Mean Variance Estimator	17
Exercise 15: That Variance Estimator Is Efficient	18
Exercise 16: The Information Identity	19
Exercise 17: Maximum Likelihood for the Exponential Rate	21
Exercise 18: Maximum Likelihood Is Invariant Under Reparameterization	22
Exercise 19: Maximum Likelihood for the Geometric	23
Exercise 20: Estimating p from Waiting Times to a Four	24
Exercise 21: Maximum Likelihood for a Power-Law Density	25
Exercise 22: Maximum Likelihood for a Normal Scale	26
Exercise 23: The Same Estimator, Reparameterized to the Variance	27
Exercise 24: Normal Estimates from Fifteen Observations	28
Exercise 25: Estimates from a Simulated Standard Normal Sample	29

Exercise 26: The Second-Derivative Test for the Bernoulli	30
Exercise 27: The Estimator of Exercise 23 Is Efficient	31
Exercise 28: Maximum Likelihood at the Boundary—the Uniform	32
Exercise 29: The Best Multiple of the Known-Mean Variance Estimator	33
Exercise 30: Method of Moments for the Uniform	35
Exercise 31: A Simulated Uniform Sample	36
Exercise 32: Method of Moments for a Rayleigh Density	37
Exercise 33: Method of Moments for Both Binomial Parameters	39
Exercise 34: Estimating p and n from Coin-Tossing Data	40
Exercise 35: Mean and Variance of the Gamma	41
Exercise 36: Why the Method of Moments Fails for the Cauchy	42
Exercise 37: Sample Moments Are Unbiased for Theoretical Moments	43
Exercise 38: Estimating an Expectation With and Without the Model	44
Exercise 39: A Confidence Interval for a Normal Mean	46
Exercise 40: Halving the Interval Costs Four Times the Sample	47
Exercise 41: An 80% Interval from a Simulated Sample	48
Exercise 42: A Confidence Interval for the Variance, Mean Known	49
Exercise 43: An Interval for σ , Mean Known	50
Exercise 44: The Same Interval With the Mean Unknown	51
Exercise 45: A 95% Interval for the Variance from Data	52
Exercise 46: Sample Size for a Short Interval	53
Exercise 47: A Large-Sample Interval for a Poisson Mean	54
Exercise 48: A Large-Sample Interval for a Bernoulli p	55
Exercise 49: A Coin-Tossing Interval for p	56
Exercise 50: A Die-Tossing Interval for p	57
Exercise 51: A Large-Sample Interval for an Exponential Mean	58
Exercise 52: Bayes Estimator for a Bernoulli with a Beta(2,2) Prior	59
Exercise 53: A Bayes Computation Where the Likelihood Cancels the Prior	60

Exercise 54: Bayes Estimator for the Poisson Mean	62
Exercise 55: Bayes Estimator with a Truncated Prior	63
Exercise 56: Minimax and Bayes Within Two Estimators	65
Exercise 57: Absolute-Error Risk for Two Estimators	67
Exercise 58: A Conjugate Gamma Bayes Estimator	69
Exercise 59: The Best Predictor and the Best Linear Predictor	70
Exercise 60: A Curved Best Predictor and Its Linear Approximation	72
Exercise 61: Sequential Updating Equals Batch Updating	74
References	76

HPS, Chapter 2: Exercises 1–61

Exercise 1

Given $f(x \mid \theta) = \frac{e^{-x^2/2\theta}}{\sqrt{2\pi\theta}}$, show that $d(X) = \frac{\sum_{i=1}^n X_i^2}{n}$, where $X_1, \dots, X_n$ is a random sample from $f(x \mid \theta)$, is an unbiased estimator of θ .

SOLUTION:

The density is normal with mean 0 and variance θ . Because the mean is zero, the second moment is the variance itself:

$$\mathbb{E}^\theta\{X_i^2\} = \text{Var}^\theta(X_i) + (\mathbb{E}^\theta\{X_i\})^2 = \theta + 0 = \theta.$$

Linearity then finishes it, with no cancellation to arrange:

$$\mathbb{E}^\theta\{d\} = \frac{1}{n} \sum_{i=1}^n \mathbb{E}^\theta\{X_i^2\} = \frac{1}{n} \cdot n\theta = \boxed{\theta}$$

for every θ , so d is unbiased. The known-mean assumption is what makes the raw second moment estimate the variance directly; Exercise 14 computes this same estimator's variance, and Exercise 15 shows it attains the bound. ■

Exercise 2

Given $f(x \mid \theta) = 1/\theta$, $0 \leq x \leq \theta$, determine c so that $d(X) = cX$ will be an unbiased estimator of θ .

SOLUTION:

On $[0, \theta]$ the mean of the uniform density is the midpoint, $E^\theta\{X\} = \theta/2$. Hence $E^\theta\{cX\} = c\theta/2$, and this equals θ for every θ exactly when

$$\boxed{c = 2} \quad \text{so } d(X) = 2X.$$

Doubling the observation corrects for the fact that a single draw sits, on average, halfway up the interval. This unbiased rule is the method-of-moments estimator of Exercise 30 as well; Exercise 28 finds the very different maximum likelihood estimator $\max_i X_i$ for the same density. ■

Exercise 3

Given $f(x \mid \theta) = \frac{x^{\theta-1}e^{-x}}{\Gamma(\theta)}$, $x > 0$, $\theta > 0$, (a) find a value of c such that $d(X) = cX$ will be an unbiased estimator of θ ; (b) determine whether it is possible to find a value of c such that $d(X) = cX^2$ will be an unbiased estimator of θ .

SOLUTION:

This is the Gamma density with shape θ and unit scale, whose moments are $E^\theta\{X\} = \theta$ and $\text{Var}^\theta(X) = \theta$, so $E^\theta\{X^2\} = \theta + \theta^2$.

(a) $E^\theta\{cX\} = c\theta$, equal to θ for all θ exactly when

$$\boxed{c = 1} \quad \text{so } d(X) = X.$$

(b) $E^\theta\{cX^2\} = c(\theta + \theta^2)$. For this to equal θ for every θ we would need $c = \theta/(\theta + \theta^2) = 1/(1 + \theta)$, which is not a constant—it depends on the very quantity being estimated. So

$$\boxed{\text{no constant } c \text{ makes } cX^2 \text{ unbiased for } \theta.}$$

The contrast is the lesson: a linear statistic can be scaled to unbiasedness because its mean is proportional to θ , but the second moment mixes θ and θ^2 , and no single multiplier can strip out the quadratic term uniformly in θ . ■

Exercise 4

Determine the restrictions that are needed on the a 's so that $Z = \sum_{i=1}^n a_i X_i$ will be an unbiased estimator of EX , where $X_1, \dots, X_n$ represents a random sample of X .

SOLUTION:

Write $\mu = E\{X\}$. By linearity, and because each X_i has the common mean μ ,

$$E\{Z\} = \sum_{i=1}^n a_i E\{X_i\} = \mu \sum_{i=1}^n a_i.$$

This equals μ for *every* value of μ exactly when the coefficient of μ is 1:

$$\sum_{i=1}^n a_i = 1.$$

That single restriction is the whole content of unbiasedness for a linear estimator; nothing about the variance enters yet. Exercise 9 takes the same weights and asks which choice, among all that satisfy $\sum a_i = 1$, has the smallest variance. ■

Exercise 5

If the bias of an estimator is denoted by $b(\theta)$ and is defined by the relation $Ed(X) = \theta + b(\theta)$, calculate the bias in the minimax estimator $d'(X) = (X + \sqrt{n}/2)/(n + \sqrt{n})$ associated with Figure 1, where X is a binomial variable with parameters n and p .

SOLUTION:

The parameter is p , and X binomial(n, p) has $E\{X\} = np$. Hence

$$E\{d'\} = \frac{E\{X\} + \sqrt{n}/2}{n + \sqrt{n}} = \frac{np + \sqrt{n}/2}{n + \sqrt{n}},$$

and the bias $b(p) = E\{d'\} - p$ is

$$b(p) = \frac{np + \sqrt{n}/2 - p(n + \sqrt{n})}{n + \sqrt{n}} = \frac{\sqrt{n}/2 - p\sqrt{n}}{n + \sqrt{n}} = \frac{\sqrt{n}(\frac{1}{2} - p)}{\sqrt{n}(\sqrt{n} + 1)} = \boxed{\frac{\frac{1}{2} - p}{\sqrt{n} + 1}}$$

using $n + \sqrt{n} = \sqrt{n}(\sqrt{n} + 1)$. The bias is zero at $p = \frac{1}{2}$ and grows linearly as p moves away from the middle, but it is damped by the factor $1/(\sqrt{n} + 1)$ and vanishes as $n \rightarrow \infty$. This estimator pulls the observed rate X/n toward $\frac{1}{2}$; the pull is exactly what buys the flat risk of Exercise 6. ■

Exercise 6

Show that the value of $\mathcal{R}(p, d') = E(d' - p)^2$, where d' is the minimax estimator of Exercise 5, is given by $1/(4(\sqrt{n} + 1)^2)$.

SOLUTION:

Decompose the mean squared error into variance and squared bias. From $d' = (X + \sqrt{n}/2)/(n + \sqrt{n})$,

$$\text{Var}(d') = \frac{\text{Var}(X)}{(n + \sqrt{n})^2} = \frac{np(1-p)}{n(\sqrt{n} + 1)^2} = \frac{p(1-p)}{(\sqrt{n} + 1)^2},$$

since $(n + \sqrt{n})^2 = n(\sqrt{n} + 1)^2$, and the squared bias is $b(p)^2 = (\frac{1}{2} - p)^2/(\sqrt{n} + 1)^2$ from Exercise 5. Adding,

$$\mathcal{R}(p, d') = \frac{p(1-p) + (\frac{1}{2} - p)^2}{(\sqrt{n} + 1)^2}.$$

The numerator collapses: $p(1-p) + (\frac{1}{2} - p)^2 = p - p^2 + \frac{1}{4} - p + p^2 = \frac{1}{4}$. Therefore

$$\boxed{\mathcal{R}(p, d') = \frac{1}{4(\sqrt{n} + 1)^2}}$$

free of p . A constant risk is what makes d' minimax: no value of p is worse than any other, so the estimator cannot be improved in its worst case, and it is the equalizer rule for this problem—the same phenomenon met in Assignment 4, here in the book's own Figure 1. ■

Exercise 7

Fish are caught in a lake until n fish of a certain type have been obtained. Let X denote the total number of fish caught at that point. Assume that the lake is so large that the extraction of these fish does not change the proportion p of this type of fish in the lake. Show that the random variable X possesses the density function given by $\binom{x-1}{n-1} p^n (1-p)^{x-n}$, $x = n, n+1, \dots$, where n is assumed to be a fixed integer. Use this result to show that $(n-1)/(X-1)$ is an unbiased estimator of p . Note that this estimator differs from the intuitive estimator n/X , which is biased for this problem.

SOLUTION:

The density. Catching continues until the n th fish of the type appears, and $X = x$ means the x th fish caught is that n th one. Two independent facts must hold: the x th catch is of the type—probability p —and among the first $x-1$ catches exactly $n-1$ are of the type—probability $\binom{x-1}{n-1} p^{n-1} (1-p)^{(x-1)-(n-1)}$. Multiplying,

$$P(X = x) = \binom{x-1}{n-1} p^n (1-p)^{x-n}, \quad x = n, n+1, \dots,$$

the negative binomial (number of trials to the n th success).

Unbiasedness of $(n-1)/(X-1)$. The key is a factorial identity that lowers both indices by one:

$$\frac{n-1}{x-1} \binom{x-1}{n-1} = \frac{n-1}{x-1} \cdot \frac{(x-1)!}{(n-1)!(x-n)!} = \frac{(x-2)!}{(n-2)!(x-n)!} = \binom{x-2}{n-2}.$$

Therefore

$$\mathbb{E}\left[\frac{n-1}{X-1}\right] = \sum_{x=n}^{\infty} \binom{x-2}{n-2} p^n (1-p)^{x-n} = p \sum_{x=n}^{\infty} \binom{x-2}{(n-1)-1} p^{n-1} (1-p)^{(x-1)-(n-1)}.$$

The sum on the right is the total probability of a negative binomial waiting for $n-1$ successes (index $x-1$ running over $n-1, n, \dots$), hence equals 1. So

$$\boxed{\mathbb{E}\left[\frac{n-1}{X-1}\right] = p}$$

The unbiased estimator is not the tempting n/X : replacing n by $n-1$ and X by $X-1$ is precisely the correction that makes the expectation come out to p , and by Jensen's inequality the convex map $x \mapsto n/x$ leaves $\mathbb{E}\{n/X\} > n/\mathbb{E}\{X\} = p$, so n/X is biased upward. This is the “inverse sampling” estimator, the natural counterpart to the binomial's X/n when it is the number of successes, not the number of trials, that is fixed in advance. ■

Exercise 8

Show that $d = \sum_{i=1}^n X_i/n$ is an efficient unbiased estimator of the parameter θ for the Poisson density $f(x | \theta) = e^{-\theta}\theta^x/x!$, $x = 0, 1, \dots$, where $X_1, \dots, X_n$ represents a random sample of X .

SOLUTION:

Unbiasedness first. Each X_i has mean θ , so by linearity $E^\theta\{d\} = \frac{1}{n} \sum_i E^\theta\{X_i\} = \theta$ for every θ .

For efficiency, compute the quantity in the bound. From $\log f(x | \theta) = -\theta + x \log \theta - \log x!$,

$$\frac{\partial \log f(x | \theta)}{\partial \theta} = -1 + \frac{x}{\theta} = \frac{x - \theta}{\theta},$$

and since $E^\theta\{(X - \theta)^2\} = \text{Var}^\theta(X) = \theta$ for the Poisson,

$$E \left[\left(\frac{\partial \log f(X | \theta)}{\partial \theta} \right)^2 \right] = \frac{\text{Var}^\theta(X)}{\theta^2} = \frac{1}{\theta}, \quad \text{so} \quad B(\theta) = \frac{1}{n \cdot (1/\theta)} = \frac{\theta}{n}.$$

The estimator's variance is $\text{Var}^\theta(d) = \text{Var}^\theta(X)/n = \theta/n$, because the X_i are independent and variances scale by $1/n^2$ across the sum of n of them. Hence

$$e(d) = \frac{B(\theta)}{\text{Var}^\theta(d)} = \frac{\theta/n}{\theta/n} = 1$$

and d is an efficient unbiased estimator in the book's sense of Definition 2 (p. 17): it attains the lower bound at every θ . This closes a small gap in the text itself, whose worked examples in §2.2 are the normal and the Bernoulli only; the Poisson computation ran through Assignments 5 and 6 and lands here as the missing third example. ■

Exercise 9

(a) What is the best set of a 's to choose in Exercise 4 if $Z = \sum_{i=1}^n a_i X_i$ is to be an unbiased estimator of EX with minimum variance? (b) What a 's should be chosen if the X_i are independent variables with the same mean but different known variances σ_i^2 ?

SOLUTION:

Exercise 4 supplies the constraint $\sum_{i=1}^n a_i = 1$: with $\mu = E\{X\}$, linearity gives $E\{Z\} = \mu \sum_i a_i$, so Z is unbiased for every μ exactly when the weights sum to one.

(a) With a common variance σ^2 and independence, $\text{Var}(Z) = \sigma^2 \sum_i a_i^2$, so the problem is to minimize $\sum_i a_i^2$ subject to $\sum_i a_i = 1$. No calculus is needed: expand the sum of squared deviations from $1/n$,

$$\sum_{i=1}^n \left(a_i - \frac{1}{n}\right)^2 = \sum_i a_i^2 - \frac{2}{n} \sum_i a_i + n \cdot \frac{1}{n^2} = \sum_i a_i^2 - \frac{1}{n},$$

using the constraint in the middle term. The left side is nonnegative and vanishes only when every $a_i = 1/n$, so $\sum_i a_i^2 \geq 1/n$ with equality exactly there:

$$a_i = \frac{1}{n} \text{ for every } i: \text{ the sample mean, with variance } \sigma^2/n$$

(b) Now $\text{Var}(Z) = \sum_i a_i^2 \sigma_i^2$, to be minimized under the same constraint. The Lagrange condition sets $\partial/\partial a_i [\sum_j a_j^2 \sigma_j^2 - \lambda \sum_j a_j] = 2a_i \sigma_i^2 - \lambda = 0$, so $a_i = \lambda/(2\sigma_i^2)$: the weights must be proportional to the reciprocal variances. The constraint fixes the constant:

$$a_i = \frac{1/\sigma_i^2}{\sum_{j=1}^n 1/\sigma_j^2} \quad \text{with} \quad \text{Var}(Z) = \left(\sum_j 1/\sigma_j^2\right)^{-1},$$

the variance following on substituting the weights: $\sum_i a_i^2 \sigma_i^2 = \sum_i \sigma_i^{-2} / (\sum_j \sigma_j^{-2})^2 = 1 / \sum_j \sigma_j^{-2}$. (That the critical point is the minimum follows because the objective is a positive-definite quadratic on the constraint plane.) Precise observations get large weights, noisy ones small; precisions add in the denominator. This is the same arithmetic that set the posterior weights in Assignment 7's normal problem, with data replacing the prior as the second voice. ■

Exercise 10

Show that the estimator $Z = (X_1 + X_2)/2$ is an unbiased estimator of EX , where $X_1, \dots, X_n$ represents a random sample of X . Calculate the efficiency of Z if X possesses the density

$$f(x | \theta) = \frac{\exp [-(1/2)(x - \theta)^2]}{\sqrt{2\pi}}.$$

SOLUTION:

Unbiasedness holds for any parent distribution: $E\{Z\} = \frac{1}{2}(E\{X_1\} + E\{X_2\}) = EX$ by linearity.

For the normal density with unit variance, $\text{Var}(Z) = \frac{1}{4}(\text{Var}(X_1) + \text{Var}(X_2)) = \frac{1}{4} \cdot 2 = \frac{1}{2}$, the two observations being independent. The bound uses

$$\log f(x | \theta) = -\frac{1}{2}(x - \theta)^2 - \frac{1}{2} \log(2\pi), \quad \frac{\partial \log f}{\partial \theta} = x - \theta, \quad E[(X - \theta)^2] = 1,$$

so $B(\theta) = 1/(n \cdot 1) = 1/n$, and Definition 2 gives

$$e(Z) = \frac{B(\theta)}{\text{Var}(Z)} = \frac{1/n}{1/2} = \frac{2}{n}$$

The number reads as an accounting of waste: Z throws away $n-2$ of the n observations, and its efficiency is exactly the fraction of the sample it uses. At $n = 2$ it is the sample mean and the efficiency is 1; for larger n an unbiased estimator with variance $1/n$ exists (the full mean), and Z delivers only $2/n$ of that precision. ■

Exercise 11

Let X be a random variable with finite second moment. Show that $E(X - b)^2$, where b is a constant, is minimized by $b = EX$.

SOLUTION:

Write $\mu = EX$, add and subtract it inside the square, and expand:

$$\begin{aligned} E\{(X - b)^2\} &= E\{[(X - \mu) + (\mu - b)]^2\} && \text{(add and subtract } \mu) \\ &= E\{(X - \mu)^2\} + 2(\mu - b) E\{X - \mu\} + (\mu - b)^2 \\ &&& \text{(expand; } \mu - b \text{ is constant)} \\ &= \text{Var}(X) + (\mu - b)^2. && (E\{X - \mu\} = 0) \end{aligned}$$

The first term does not involve b and the second is nonnegative, so the whole expression is minimized exactly when $(\mu - b)^2 = 0$:

$$b = EX, \text{ with minimum value } \text{Var}(X)$$

The finite second moment is what makes every expectation above exist. Small as it is, this fact has been carrying weight all course: it is the reason the best constant predictor is the mean (notes, Thm. 2.8.1), the inner step of the posterior-mean theorem used throughout Assignment 7, and—applied conditionally—the reason the best predictor given data is the conditional expectation, which Exercises 59 and 60 below put to work. ■

Exercise 12

Find the lower bound of the variance for unbiased estimators of the parameter θ for the

Cauchy density $f(x | \theta) = \frac{1}{\pi[1 + (x - \theta)^2]}$.

SOLUTION:

The bound is $B(\theta) = 1/\{n \text{E}[(\partial \log f / \partial \theta)^2]\}$. From $\log f = -\log \pi - \log[1 + (x - \theta)^2]$,

$$\frac{\partial \log f}{\partial \theta} = \frac{2(x - \theta)}{1 + (x - \theta)^2}, \quad \left(\frac{\partial \log f}{\partial \theta}\right)^2 = \frac{4(x - \theta)^2}{[1 + (x - \theta)^2]^2}.$$

Averaging against f and substituting $u = x - \theta$,

$$\text{E}\left[\left(\frac{\partial \log f}{\partial \theta}\right)^2\right] = \int_{-\infty}^{\infty} \frac{4(x - \theta)^2}{[1 + (x - \theta)^2]^2} \cdot \frac{dx}{\pi[1 + (x - \theta)^2]} = \frac{4}{\pi} \int_{-\infty}^{\infty} \frac{u^2}{(1 + u^2)^3} du.$$

The integral is a Beta integral: $\int_{-\infty}^{\infty} u^2(1 + u^2)^{-3} du = B(\frac{3}{2}, \frac{3}{2}) = \Gamma(\frac{3}{2})^2/\Gamma(3) = (\sqrt{\pi}/2)^2/2 = \pi/8$. Hence the information per observation is $(4/\pi)(\pi/8) = \frac{1}{2}$, and

$$B(\theta) = \frac{1}{n \cdot \frac{1}{2}} = \frac{2}{n}$$

The bound is free of θ , as a pure location problem must be, and equals twice $1/n$: locating a Cauchy center is worth only half a normal observation apiece, the heavy tails costing information. Contrast Exercise 36, where the same density defeats the method of moments entirely because its mean does not exist—the Fisher information is finite even though no moment is. ■

Exercise 13

Why is the derivation that produced the inequality in Theorem 1 not applicable to the problem of estimating θ for the uniform density $f(x | \theta) = 1/\theta$, $0 \leq x \leq \theta$?

SOLUTION:

The derivation of Theorem 1 differentiates the identity $\int f(x | \theta) dx = 1$ under the integral sign to conclude that the score has mean zero,

$$\int \frac{\partial \log f(x | \theta)}{\partial \theta} f(x | \theta) dx = 0 \quad (\text{Formula (5)}),$$

and this zero-mean fact is what the Cauchy–Schwarz step then leverages into the variance inequality. Interchanging $\partial/\partial\theta$ with $\int$ is legitimate only when the region of integration does not itself depend on θ —one of the regularity conditions the theorem quietly assumes.

For the uniform on $[0, \theta]$ the support *is* the parameter. Here $\int_0^\theta (1/\theta) dx = 1$, and differentiating the two sides gives, by Leibniz’s rule with a moving upper limit,

$$\frac{d}{d\theta} \int_0^\theta \frac{1}{\theta} dx = \underbrace{\frac{1}{\theta} \Big|_{x=\theta}}_{\text{boundary term}} + \int_0^\theta \frac{\partial}{\partial \theta} \frac{1}{\theta} dx = \frac{1}{\theta} - \frac{1}{\theta} = 0,$$

but the naive “differentiate inside” step keeps only the integral piece $\int_0^\theta (-1/\theta^2) dx = -1/\theta \neq 0$. The boundary term the moving endpoint contributes is exactly what the interchange drops, so Formula (5) fails and with it the whole derivation. This is no mere technicality: the best unbiased estimator of θ here, a multiple of $\max_i X_i$ (Exercise 28), has variance of order $1/n^2$, far below any $1/n$ bound the theorem would have predicted. The regularity conditions are what separate the smooth families of

Exercises 8–12 from this one.



Exercise 14

Calculate the mean and variance of the estimator $Z = \frac{\sum_{i=1}^n (X_i - \mu)^2}{n}$ for the parameter θ in

$$f(x \mid \theta) = \frac{e^{-[(x-\mu)^2/2\theta]}}{\sqrt{2\pi\theta}}.$$

Use the fact from Volume I that $(X - \mu)^2/\theta$ possesses a chi-square distribution with one degree of freedom and hence that nZ/θ possesses a chi-square distribution with n degrees of freedom. This assumes that $X_1, \dots, X_n$ represent a random sample of X .

SOLUTION:

Here θ is the variance of a normal with known mean μ . Following the given fact, write $W = nZ/\theta$, so $W \sim \chi_n^2$ with $E\{W\} = n$ and $\text{Var}(W) = 2n$, and $Z = \theta W/n$. Then

$$E^\theta\{Z\} = \frac{\theta}{n} E\{W\} = \frac{\theta}{n} \cdot n = \boxed{\theta} \quad (\text{linearity; } \chi_n^2 \text{ mean})$$

$$\text{Var}^\theta(Z) = \frac{\theta^2}{n^2} \text{Var}(W) = \frac{\theta^2}{n^2} \cdot 2n = \boxed{\frac{2\theta^2}{n}} \quad (\text{scaling squares; } \chi_n^2 \text{ variance})$$

So Z is unbiased for the variance with sampling variance $2\theta^2/n$. Exercise 15 checks this against the bound and finds efficiency 1, and Exercise 29 revisits this same Z to ask the sharper question of whether unbiasedness is even worth insisting on. ■

Exercise 15

Use the result of Exercise 14 to show that Z as defined there is an efficient unbiased estimator of θ .

SOLUTION:

Exercise 14 gives $E^\theta\{Z\} = \theta$ and $\text{Var}^\theta(Z) = 2\theta^2/n$. It remains to find the bound $B(\theta)$ for this model, in which θ is the variance. The second-derivative form of the information (Exercise 16) is the cleaner route. From

$$\log f(x \mid \theta) = -\frac{(x - \mu)^2}{2\theta} - \frac{1}{2} \log \theta - \frac{1}{2} \log 2\pi,$$

$$\frac{\partial \log f}{\partial \theta} = \frac{(x - \mu)^2}{2\theta^2} - \frac{1}{2\theta}, \quad \frac{\partial^2 \log f}{\partial \theta^2} = -\frac{(x - \mu)^2}{\theta^3} + \frac{1}{2\theta^2}.$$

Taking expectations and using $E^\theta\{(X - \mu)^2\} = \theta$,

$$E\left[\left(\frac{\partial \log f}{\partial \theta}\right)^2\right] = -E\left[\frac{\partial^2 \log f}{\partial \theta^2}\right] = \frac{\theta}{\theta^3} - \frac{1}{2\theta^2} = \frac{1}{2\theta^2}, \quad \text{so} \quad B(\theta) = \frac{1}{n \cdot \frac{1}{2\theta^2}} = \frac{2\theta^2}{n}.$$

Therefore

$$e(Z) = \frac{B(\theta)}{\text{Var}^\theta(Z)} = \frac{2\theta^2/n}{2\theta^2/n} = 1$$

and Z attains the bound at every θ : it is efficient. With $\mu = 0$ this is exactly the estimator of Exercise 1, and—read as an estimator of a variance $\varphi = \theta^2$ —it reappears as the maximum likelihood estimator of Exercise 23, which Exercise 27 pronounces efficient on the strength of this very computation. ■

Exercise 16

By differentiating (5) under the integral sign with respect to θ and using the relationship

$$\frac{\partial L}{\partial \theta} = L \frac{\partial \log L}{\partial \theta},$$

show that

$$E \left(\frac{\partial \log f(X | \theta)}{\partial \theta} \right)^2 = -E \left(\frac{\partial^2 \log f(X | \theta)}{\partial \theta^2} \right).$$

This gives an alternative method for calculating the lower bound in (2) when it is applicable.

SOLUTION:

The book's display (5) (p. 19) is the statement that the derivative of the log likelihood has mean zero, written for a single observation as

$$\int \frac{\partial \log f(x | \theta)}{\partial \theta} f(x | \theta) dx = 0 \quad \text{for every } \theta,$$

which itself came from differentiating $\int f dx = 1$ under the integral sign. Differentiate it once more in θ , the product rule acting on the two θ -dependent factors:

$$\begin{aligned} 0 &= \int \frac{\partial^2 \log f(x | \theta)}{\partial \theta^2} f(x | \theta) dx + \int \frac{\partial \log f(x | \theta)}{\partial \theta} \frac{\partial f(x | \theta)}{\partial \theta} dx \\ &\quad \text{(product rule under the integral)} \\ &= \int \frac{\partial^2 \log f}{\partial \theta^2} f dx + \int \left(\frac{\partial \log f}{\partial \theta} \right)^2 f dx \\ &\quad \text{(the given relation, } \partial f / \partial \theta = f \partial \log f / \partial \theta) \end{aligned}$$

$$= \mathbb{E} \left(\frac{\partial^2 \log f(X | \theta)}{\partial \theta^2} \right) + \mathbb{E} \left(\frac{\partial \log f(X | \theta)}{\partial \theta} \right)^2,$$

(both integrals are expectations against f)

and rearranging gives the display:

$$\mathbb{E} \left(\frac{\partial \log f(X | \theta)}{\partial \theta} \right)^2 = - \mathbb{E} \left(\frac{\partial^2 \log f(X | \theta)}{\partial \theta^2} \right)$$

The same computation runs verbatim with the joint likelihood L in place of f , and with sums in place of integrals for discrete families. This is the identity the course text never states in its own §2.2—its bound (2) is defined through the squared first derivative alone—and Assignment 5 verified it model by model; here it arrives in general, from two differentiations of the normalization. DeGroot and Schervish prove it as Theorem 8.8.1 (pp. 515–516). The practical value is as the exercise says: second derivatives are often easier to average than squared first derivatives, since cross terms die—exactly the shortcut used in Exercises 12 and 15. ■

Exercise 17

Find the maximum likelihood estimator of θ for $f(x | \theta) = \theta e^{-\theta x}$, $x > 0$, based on a random sample of size n .

SOLUTION:

The likelihood and its logarithm are

$$L(\theta) = \prod_{i=1}^n \theta e^{-\theta x_i} = \theta^n e^{-\theta \sum_{i=1}^n x_i}, \quad \log L(\theta) = n \log \theta - \theta \sum_{i=1}^n x_i.$$

Differentiating and setting to zero,

$$\frac{\partial \log L}{\partial \theta} = \frac{n}{\theta} - \sum_i x_i = 0 \implies \boxed{\hat{\theta} = \frac{n}{\sum_i X_i} = \frac{1}{\bar{X}}}$$

The second derivative $-n/\theta^2 < 0$ confirms a maximum. The estimator is the reciprocal of the sample mean, which is exactly right for a rate parameter: $E\{X\} = 1/\theta$, so a large average of waiting times argues for a small rate. Exercise 18 estimates the mean $1/\theta$ itself and recovers $\bar{X}$. ■

Exercise 18

Given the information of Exercise 17, find the maximum likelihood estimator of EX and compare it with the result obtained in Exercise 17.

SOLUTION:

For the exponential $f(x | \theta) = \theta e^{-\theta x}$ the mean is $EX = 1/\theta$. The maximum likelihood estimator of a one-to-one function of a parameter is that function of the estimator (the invariance property), so with $\hat{\theta} = 1/\bar{X}$ from Exercise 17,

$$\widehat{EX} = \frac{1}{\hat{\theta}} = \bar{X}$$

The comparison is the point: Exercise 17 estimates the rate by $1/\bar{X}$, and estimating the mean $1/\theta$ simply inverts it back to $\bar{X}$. One need not redo the maximization in the new parameterization—the likelihood is a function of the data, not of the name given to θ , so its maximizer transforms along with the parameter. The mean of an exponential is estimated by the plainest statistic there is, the sample average. ■

Exercise 19

Find the maximum likelihood estimator of p for the geometric density $f(x | p) = p(1 - p)^x$, $x = 0, 1, 2, \dots$, based on a random sample of size n .

SOLUTION:

With $s = \sum_i x_i$, the likelihood and log likelihood are

$$L(p) = \prod_{i=1}^n p(1 - p)^{x_i} = p^n (1 - p)^s, \quad \log L(p) = n \log p + s \log(1 - p).$$

Setting the derivative to zero,

$$\frac{\partial \log L}{\partial p} = \frac{n}{p} - \frac{s}{1 - p} = 0 \implies n(1 - p) = ps \implies n = p(n + s)$$

$$\implies \boxed{\hat{p} = \frac{n}{n + \sum_i X_i} = \frac{1}{1 + \bar{X}}}$$

and $\partial^2 \log L / \partial p^2 = -n/p^2 - s/(1 - p)^2 < 0$ confirms the maximum. Since $E\{X\} = (1 - p)/p$ counts the failures before the first success, $1/(1 + \bar{X})$ is the natural estimate: more failures per trial on average means a smaller success probability. Exercise 20 applies this to twenty rounds of rolling a die until a four appears. ■

Exercise 20

Perform the following experiment twenty times. Roll a die until a four spot is obtained and record the number of rolls that were required. Let $x_1, \dots, x_{20}$ represent the experimental outcomes in 20 such experiments. Use these values and the result in Exercise 19 to estimate the value of p , which is $1/6$, for this problem.

SOLUTION:

The number of rolls N_i required to see the first four is a geometric *count of trials*, $N_i \geq 1$, whereas Exercise 19's variable $X = N - 1$ counts the failures before it. Its estimator $\hat{p} = 1/(1 + \bar{X})$ therefore becomes, in terms of the recorded counts,

$$\hat{p} = \frac{1}{1 + (\bar{N} - 1)} = \frac{1}{\bar{N}},$$

the reciprocal of the average waiting time—sensible, since $E\{N\} = 1/p$. Twenty simulated rounds (seeded generator, standing in for physical die rolls) gave

$N : 3, 10, 6, 8, 2, 31, 1, 4, 2, 1, 6, 4, 1, 10, 2, 4, 2, 6, 3, 2,$

with $\sum N_i = 108$ and $\bar{N} = 108/20 = 5.4$. Hence

$$\hat{p} = \frac{1}{\bar{N}} = \frac{1}{5.4} = 0.185$$

against the true $p = 1/6 \approx 0.167$. The lone value $N = 31$ is a reminder of the geometric's long right tail: waiting times occasionally run far past the mean of 6, and with only twenty rounds one such run pulls $\bar{N}$ and hence $\hat{p}$ around noticeably. ■

Exercise 21

Find the maximum likelihood estimator of θ , based on a random sample of size n , for the density $f(x \mid \theta) = (1 + \theta)x^\theta$, $0 \leq x \leq 1$, $\theta > 0$.

SOLUTION:

The likelihood and its logarithm are

$$L(\theta) = (1 + \theta)^n \left(\prod_i x_i \right)^\theta, \quad \log L(\theta) = n \log(1 + \theta) + \theta \sum_{i=1}^n \log x_i.$$

Setting the derivative to zero,

$$\frac{\partial \log L}{\partial \theta} = \frac{n}{1 + \theta} + \sum_i \log x_i = 0 \implies 1 + \theta = \frac{-n}{\sum_i \log x_i} \implies \boxed{\hat{\theta} = -\frac{n}{\sum_{i=1}^n \log X_i} - 1}$$

Because each $x_i \in (0, 1)$, $\sum_i \log x_i < 0$ and the estimator is positive, as θ must be. The second derivative $-n/(1 + \theta)^2 < 0$ confirms a maximum. The sufficient statistic is $\sum \log x_i$: data bunched near 1 (small $|\sum \log x_i|$) drive $\hat{\theta}$ large, matching the fact that large θ tilts the density's mass toward the upper endpoint. ■

Exercise 22

Given $f(x \mid \theta) = \frac{e^{-(x^2/2\theta^2)}}{\sqrt{2\pi}\theta}$, find the maximum likelihood estimator of θ based on a random sample of size n .

SOLUTION:

This is the normal density with mean 0 and standard deviation θ (variance θ^2). Its log and the log likelihood are

$$\log f(x \mid \theta) = -\frac{x^2}{2\theta^2} - \log \theta - \frac{1}{2} \log 2\pi, \quad \log L(\theta) = -\frac{\sum_i x_i^2}{2\theta^2} - n \log \theta - \frac{n}{2} \log 2\pi.$$

Differentiating in θ and setting to zero,

$$\frac{\partial \log L}{\partial \theta} = \frac{\sum_i x_i^2}{\theta^3} - \frac{n}{\theta} = 0 \implies \sum_i x_i^2 = n\theta^2 \implies \boxed{\hat{\theta} = \sqrt{\frac{\sum_{i=1}^n X_i^2}{n}}}$$

the root-mean-square of the observations, which is exactly right for a scale measured about a known center of zero. Exercise 23 reparameterizes to the variance $\varphi = \theta^2$ and Exercise 27 shows the resulting estimator is efficient and unbiased. ■

Exercise 23

Treat $\varphi = \theta^2$ as the parameter in Exercise 22 and then find the maximum likelihood estimator of φ . Compare your result with the result obtained in Exercise 22.

SOLUTION:

By the invariance of maximum likelihood under reparameterization, the estimator of $\varphi = \theta^2$ is the square of the estimator of θ :

$$\hat{\varphi} = \hat{\theta}^2 = \frac{\sum_{i=1}^n X_i^2}{n}$$

One can also see it directly: writing the density in terms of φ gives $\log L = -\sum_i x_i^2/(2\varphi) - \frac{n}{2} \log \varphi - \frac{n}{2} \log 2\pi$, whose derivative $\sum_i x_i^2/(2\varphi^2) - n/(2\varphi)$ vanishes at $\varphi = \sum_i x_i^2/n$. Either way the answer is the same statistic squared, illustrating that maximum likelihood does not privilege a parameterization—estimating θ then squaring, or estimating φ from scratch, land on the identical number. And $\hat{\varphi} = \sum X_i^2/n$ is precisely the estimator Z of Exercises 14–15 with $\mu = 0$; Exercise 27 records the consequence. ■

Exercise 24

Given the following random sample values of a normal variable, for which $\mu = 10$ and $\sigma = 3$, use them and (10) to obtain estimates of μ and σ^2 : 12.7, 6.6, 5.6, 14.3, 11.4, 4.3, 7.2, 10.8, 13.8, 11.2, 10.0, 12.8, 7.1, 14.0, 6.1.

SOLUTION:

Formula (10) gives the maximum likelihood estimates for a normal sample, $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = \frac{1}{n} \sum_i (x_i - \bar{x})^2$. With $n = 15$ the sum of the fifteen values is 147.9, so

$$\hat{\mu} = \bar{x} = \frac{147.9}{15} = \boxed{9.86} \quad (\text{true } \mu = 10).$$

The sum of squared deviations from $\bar{x}$ is $\sum_i (x_i - \bar{x})^2 = 162.68$, hence

$$\hat{\sigma}^2 = \frac{162.68}{15} = \boxed{10.85} \quad (\text{true } \sigma^2 = 9),$$

with $\hat{\sigma} = 3.29$ against the true $\sigma = 3$. Both estimates land near the truth, the mean almost exactly and the variance a little high—unremarkable scatter for $n = 15$. Note the estimates were formed without using the known values of μ and σ ; those are quoted only so the accuracy can be judged. ■

Exercise 25

Use Table III of the appendix to obtain a random sample of size 10 of a standard normal variable. Use your sample values and (10) to obtain estimates of μ and σ , which are 0 and 1 here.

SOLUTION:

Reading the first column of Table III (HPS, p. 223), ten standard normal values are

−1.752, −0.291, −0.933, −0.450, 0.512, −0.702, 0.284, −0.509, −1.776, −0.044.

Formula (10) then estimates

$$\hat{\mu} = \bar{x} = \boxed{-0.566} \quad (\text{true } 0), \quad \hat{\sigma} = \sqrt{\frac{1}{10} \sum_i (x_i - \bar{x})^2} = \boxed{0.726} \quad (\text{true } 1).$$

With only ten observations the estimates are noticeably off—this stretch of the table happened to run negative, dragging $\hat{\mu}$ well below zero, and the spread came out narrow. This is the point of doing it by hand once: at $n = 10$ the sampling error in $\hat{\mu}$ has standard deviation $1/\sqrt{10} \approx 0.32$, so a value near -0.57 is under a two-standard-error excursion, uncomfortable but not rare. The estimates would tighten around 0 and 1 as n grew. ■

Exercise 26

Use the second derivative test to show that $\hat{p} = \sum_{i=1}^n X_i/n$ does maximize the likelihood function for the Bernoulli density discussed in Section 2.4.

SOLUTION:

With $s = \sum_i x_i$ the Bernoulli likelihood is $L(p) = p^s(1-p)^{n-s}$, so

$$\log L(p) = s \log p + (n-s) \log(1-p), \quad \frac{\partial \log L}{\partial p} = \frac{s}{p} - \frac{n-s}{1-p}.$$

Setting this to zero gives $s(1-p) = (n-s)p$, i.e. $s = np$, so the critical point is $\hat{p} = s/n = \bar{x}$. The second derivative is

$$\frac{\partial^2 \log L}{\partial p^2} = -\frac{s}{p^2} - \frac{n-s}{(1-p)^2},$$

which is strictly negative for every $p \in (0,1)$ (both terms are negative whenever $0 < s < n$). A negative second derivative at the critical point—indeed everywhere—confirms that $\hat{p} = \bar{x}$ is a maximum, not a minimum or inflection. Because $\log L$ is strictly concave on $(0,1)$, the maximum is unique. The boundary cases $s = 0$ and $s = n$ are handled separately: there $\log L$ is monotone and the maximizer is $\hat{p} = 0$ or 1 , still equal to $\bar{x}$. ■

Exercise 27

Use the results of Exercise 15 to show that the maximum likelihood estimator of Exercise 23 is an efficient unbiased estimator.

SOLUTION:

Exercise 23's maximum likelihood estimator of the variance $\varphi = \theta^2$ is $\hat{\varphi} = \sum_i X_i^2/n$. In the density of Exercise 22 the mean is known to be 0, so with $\mu = 0$ and the variance called φ ,

$$\hat{\varphi} = \frac{\sum_{i=1}^n X_i^2}{n} = \frac{\sum_{i=1}^n (X_i - \mu)^2}{n} = Z,$$

the very estimator studied in Exercises 14 and 15. Exercise 15 established, for the normal with known mean and unknown variance, that Z is unbiased for the variance and attains the variance bound—efficiency 1. Nothing in that argument used the value of μ , so it applies verbatim with $\mu = 0$. Hence

$$\hat{\varphi} = \frac{\sum_i X_i^2}{n} \text{ is an efficient unbiased estimator of } \varphi.$$

The maximum likelihood estimator here happens to coincide with the efficient unbiased one; that is a feature of this normal-scale model, not a general rule—in Exercise 5's binomial problem the minimax (and Bayes) estimator was deliberately *biased*. ■

Exercise 28

Given $f(x \mid \theta) = 1/\theta$, $0 \leq x \leq \theta$, show that for a random sample of size n the estimator $\hat{\theta}$ that maximizes the likelihood function $(1/\theta)^n$ is given by $\hat{\theta} = \max(X_1, \dots, X_n)$, that is, the largest of the sample values. This is an illustration of a problem for which the usual calculus techniques of finding a maximizing value are inappropriate.

SOLUTION:

The likelihood needs care with its support. Each factor $f(x_i \mid \theta) = 1/\theta$ requires $0 \leq x_i \leq \theta$; if any $x_i > \theta$ that factor is 0. So

$$L(\theta) = \prod_{i=1}^n f(x_i \mid \theta) = \begin{cases} \theta^{-n}, & \theta \geq \max_i x_i, \\ 0, & \theta < \max_i x_i. \end{cases}$$

For $\theta \geq \max_i x_i$ the likelihood θ^{-n} is strictly *decreasing* in θ , so it is largest at the smallest admissible θ , namely $\theta = \max_i x_i$:

$$\hat{\theta} = \max(X_1, \dots, X_n)$$

Calculus is powerless here: $\partial \log L / \partial \theta = -n/\theta$ is never zero, so setting the derivative to zero finds nothing. The maximum sits at the boundary of the allowed region, jammed against the constraint $\theta \geq \max_i x_i$, precisely because θ indexes the support—the same feature that (in Exercise 13) voids the variance bound for this density. ■

Exercise 29

Find the value of k that minimizes $E(ks^2 - \sigma^2)^2$, where

$$s^2 = \frac{\sum (X_i - \mu)^2}{n},$$

and the X_i represent a random sample of a normal variable. Use the fact that $\sum (X_i - \mu)^2 / \sigma^2$ is a chi-square variable with n degrees of freedom and that the mean and variance of such a variable are equal to n and $2n$, respectively. What does this result show about the requirement of unbiasedness if expected squared error is the basis for comparing estimators, rather than the variance?

SOLUTION:

Write $W = ns^2/\sigma^2 \sim \chi_n^2$, so $s^2 = \sigma^2 W/n$, $E\{W\} = n$, $\text{Var}(W) = 2n$, and $E\{W^2\} = \text{Var}(W) + (E\{W\})^2 = 2n + n^2$. Then

$$\begin{aligned} E\{(ks^2 - \sigma^2)^2\} &= \sigma^4 E\left\{\left(\frac{kW}{n} - 1\right)^2\right\} && \text{(factor out } \sigma^4) \\ &= \sigma^4 \left[\frac{k^2}{n^2} E\{W^2\} - \frac{2k}{n} E\{W\} + 1 \right] && \text{(expand the square; linearity)} \\ &= \sigma^4 \left[\frac{k^2(n^2 + 2n)}{n^2} - 2k + 1 \right] = \sigma^4 \left[\frac{k^2(n+2)}{n} - 2k + 1 \right]. && \text{(substitute the } \chi_n^2 \text{ moments)} \end{aligned}$$

This is a parabola in k opening upward; its derivative $2k(n+2)/n - 2$ vanishes at

$$k = \frac{n}{n+2}$$

which is therefore the minimizer, with minimized expected squared error $\sigma^4 \left[1 - \frac{n}{n+2}\right] =$

$$2\sigma^4/(n+2).$$

The point of the question is what k is *not*: it is not 1. Exercise 14's estimator s^2 (the case $k = 1$) is unbiased with expected squared error $2\sigma^4/n$, and shrinking it by the factor $n/(n+2)$ trades a small bias for a variance reduction that wins outright, at every σ^2 and every n (Figure 1). So if expected squared error is the criterion, requiring unbiasedness costs something real: the best unbiased estimator is not the best estimator, and insisting on $E\{d\} = \theta$ amounts to optimizing only the variance half of the bias–variance decomposition. The course has met this trade twice before—the shaded estimator of Assignment 3 and the shrinkage discussion of the notes' §2.10—and this exercise is its cleanest closed-form instance.

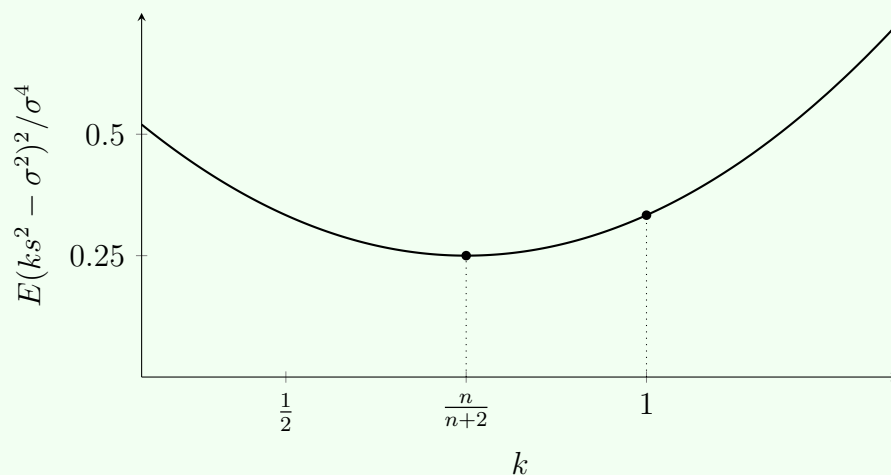


Figure 1. THE COST OF UNBIASEDNESS. Expected squared error of ks^2 as a function of k , drawn for $n = 6$ in units of σ^4 . The unbiased choice $k = 1$ (right dot, error $2/n = 0.333$) sits on the rising branch; the minimizer $k = n/(n+2) = 0.75$ (left dot, error $2/(n+2) = 0.25$) shrinks the estimator toward zero and beats it at every σ^2 .



Exercise 30

Use the method of moments to estimate the parameter θ for the uniform density $f(x \mid \theta) = 1/\theta, 0 \leq x \leq \theta$, based on a random sample of size n .

SOLUTION:

The method of moments matches the first theoretical moment to the first sample moment. The theoretical mean is $E\{X\} = \theta/2$, and the sample mean is $\bar{X}$, so setting them equal,

$$\frac{\theta}{2} = \bar{X} \implies \boxed{\hat{\theta} = 2\bar{X}}$$

This is the same estimator that unbiasedness produced from a single observation in Exercise 2, now built from the whole sample's average. It is unbiased ($E\{2\bar{X}\} = 2 \cdot \theta/2 = \theta$), but it is not the maximum likelihood estimator: Exercise 28 found $\max_i X_i$, which uses only the extreme observation yet has far smaller variance ($O(1/n^2)$ against the moment estimator's $O(1/n)$). A rare case where the plain moment estimator is badly beaten—because the density's information lives at its moving boundary, not in its average. ■

Exercise 31

Use Table II of the appendix to obtain a random sample of size 20 from the uniform density $f(x \mid \theta) = 1/\theta, 0 \leq x \leq \theta$, where $\theta = 1$. Use your sample values and the result in Exercise 30 to estimate the value of θ .

SOLUTION:

Reading the first twenty five-digit groups of Table II (HPS, p. 221) as decimal fractions (0.27767, 0.43584, 0.85301, ...) gives a uniform sample on $[0, 1]$ with mean $\bar{x} = 0.560$. The Exercise 30 estimator is then

$$\hat{\theta} = 2\bar{x} = 2(0.560) = 1.121 \quad (\text{true } \theta = 1).$$

The estimate overshoots 1 because this stretch of the table averaged a little above the population mean $\frac{1}{2}$; the moment estimator's standard deviation here is $2 \cdot \sqrt{\theta^2/12}/\sqrt{20} = \theta/\sqrt{60} \approx 0.13$, so 1.121 is within one standard error of the truth. For comparison the maximum likelihood estimator $\max_i x_i$ would have used only the sample maximum—here 0.940—which for twenty uniforms sits near $1 - 1/21 \approx 0.95$ and typically lands closer. ■

Exercise 32

Use the method of moments to estimate the parameter θ for the density $f(x \mid \theta) = \frac{x \exp[-(x^2/2\theta)]}{\theta}$, $x > 0$, $\theta > 0$, based on a random sample of size n and employing (a) the first moment, (b) the second moment.

SOLUTION:

This is a Rayleigh density with scale parameter θ ; the substitution $t = x^2/2\theta$ evaluates its moments. For the first,

$$E\{X\} = \int_0^\infty x \cdot \frac{x}{\theta} e^{-x^2/2\theta} dx = \frac{1}{\theta} \int_0^\infty x^2 e^{-x^2/2\theta} dx = \sqrt{\frac{\pi\theta}{2}},$$

and for the second, with $x dx = \theta dt$,

$$E\{X^2\} = \frac{1}{\theta} \int_0^\infty x^3 e^{-x^2/2\theta} dx = \frac{1}{\theta} \int_0^\infty 2\theta t \cdot e^{-t} \theta dt = 2\theta \int_0^\infty t e^{-t} dt = 2\theta.$$

(a) Matching $E\{X\} = \sqrt{\pi\theta/2}$ to $\bar{X}$ and solving,

$$\sqrt{\frac{\pi\theta}{2}} = \bar{X} \implies \boxed{\hat{\theta} = \frac{2\bar{X}^2}{\pi}}$$

(b) Matching $E\{X^2\} = 2\theta$ to the second sample moment $\frac{1}{n} \sum_i X_i^2$,

$$2\theta = \frac{1}{n} \sum_i X_i^2 \implies \boxed{\hat{\theta} = \frac{\sum_{i=1}^n X_i^2}{2n}}$$

Two moment equations, two estimators of the same θ —the freedom the method leaves once a distribution has more moments than parameters. The second-moment version is the tidier of the two here (no π , and $\sum X_i^2$ is the natural sufficient statistic for this

density), and it is in fact the maximum likelihood estimator; the first-moment version is a legitimate competitor but slightly less efficient. ■

Exercise 33

Use the method of moments to estimate the parameters p and n for the binomial density $f(x | \theta) = \binom{n}{x} p^x (1-p)^{n-x}$ based on the m random sample values $X_1, \dots, X_m$.

SOLUTION:

Two unknowns require two moment equations. Let $m_1 = \frac{1}{m} \sum_i X_i$ and $\hat{s}^2 = \frac{1}{m} \sum_i (X_i - m_1)^2$ be the first sample moment and the sample variance. The theoretical mean and variance of the binomial are

$$E\{X\} = np, \quad \text{Var}(X) = np(1-p).$$

Match them: $np = m_1$ and $np(1-p) = \hat{s}^2$. Dividing the second equation by the first eliminates n ,

$$1-p = \frac{\hat{s}^2}{m_1} \implies \boxed{\hat{p} = 1 - \frac{\hat{s}^2}{m_1}}$$

and then $n = m_1/p$ gives

$$\boxed{\hat{n} = \frac{m_1}{\hat{p}} = \frac{m_1^2}{m_1 - \hat{s}^2}}$$

(rounded to an integer in practice). The estimator is workable but notoriously delicate: it requires $\hat{s}^2 < m_1$ for $\hat{n}$ to be positive, and because $\hat{n}$ divides by the small difference $m_1 - \hat{s}^2$, ordinary sampling noise in $\hat{s}^2$ can throw it a long way. Exercise 34 exhibits exactly this instability on simulated data. ■

Exercise 34

Perform the following experiment twenty times. Toss a coin eight times and record the number of heads obtained. Let $x_1, \dots, x_{20}$ represent the experimental outcomes. Use those values and the result in Exercise 33 to estimate the values of p and n , which are $1/2$ and 8 here.

SOLUTION:

Twenty simulated rounds of eight tosses (seeded generator in place of a physical coin) gave the head counts

6, 5, 5, 4, 3, 4, 6, 1, 3, 7, 6, 5, 3, 1, 3, 5, 3, 4, 3, 3,

with first sample moment $m_1 = 4.0$ and sample variance $\hat{s}^2 = 2.5$. The Exercise 33 formulas then give

$$\hat{p} = 1 - \frac{\hat{s}^2}{m_1} = 1 - \frac{2.5}{4.0} = \boxed{0.375} \quad (\text{true } 1/2),$$

$$\hat{n} = \frac{m_1^2}{m_1 - \hat{s}^2} = \frac{16}{4.0 - 2.5} = \boxed{10.7} \rightarrow 11 \quad (\text{true } 8).$$

The mean is matched exactly ($\hat{n}\hat{p} = 4 = m_1$ by construction), but splitting that mean into n and p went badly: the sample variance came out a touch high (2.5 against the true $np(1-p) = 2$), and dividing by the small gap $m_1 - \hat{s}^2 = 1.5$ magnified that slip into $\hat{n} = 11$ and a correspondingly low $\hat{p} = 0.375$. This is the instability warned of in Exercise 33 made concrete—estimating a binomial’s number of trials from replicated counts is a genuinely hard problem, and the moment estimator is not up to it at $m = 20$. ■

Exercise 35

Show that the mean and variance of the gamma density

$$f(x \mid \alpha, \beta) = \frac{e^{-(x/\beta)} x^{\alpha-1}}{\beta^\alpha \Gamma(\alpha)}, \quad x > 0, \alpha > 0, \beta > 0,$$

are given by the formulas $\mu = \alpha\beta$ and $\sigma^2 = \alpha\beta^2$.

SOLUTION:

Everything follows from the normalization $\int_0^\infty x^{s-1} e^{-x/\beta} dx = \beta^s \Gamma(s)$, valid for any $s > 0$. The mean uses $s = \alpha + 1$:

$$E\{X\} = \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^\infty x^\alpha e^{-x/\beta} dx = \frac{\beta^{\alpha+1} \Gamma(\alpha+1)}{\beta^\alpha \Gamma(\alpha)} = \beta \cdot \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)} = \boxed{\alpha\beta}$$

using $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$. The second moment uses $s = \alpha + 2$:

$$E\{X^2\} = \frac{\beta^{\alpha+2} \Gamma(\alpha+2)}{\beta^\alpha \Gamma(\alpha)} = \beta^2 \cdot \frac{(\alpha+1)\alpha \Gamma(\alpha)}{\Gamma(\alpha)} = \alpha(\alpha+1)\beta^2.$$

Subtracting the squared mean,

$$\text{Var}(X) = E\{X^2\} - (E\{X\})^2 = \alpha(\alpha+1)\beta^2 - \alpha^2\beta^2 = \boxed{\alpha\beta^2}$$

The recursion $\Gamma(s+1) = s\Gamma(s)$ does all the work; no integration by parts is needed once the normalizing integral is known. These are the moments behind the Gamma prior of Assignment 7 and the conjugate posteriors of Exercises 54 and 58 below. ■

Exercise 36

Why would the method of moments fail in trying to estimate the parameter θ for the Cauchy density $f(x \mid \theta) = \frac{1}{\pi[1 + (x - \theta)^2]}$?

SOLUTION:

The method of moments begins by equating the theoretical first moment $E\{X\}$ to the sample mean $\bar{X}$. But for the Cauchy density $E\{X\}$ *does not exist*: the defining integral

$$\int_{-\infty}^{\infty} \frac{|x|}{\pi[1 + (x - \theta)^2]} dx$$

diverges, because the integrand decays only like $1/|x|$ for large $|x|$ and $\int^{\infty} dx/x = \infty$. With no finite mean there is no first-moment equation to write down, and every higher moment diverges too, so the method has nothing to match. The failure is structural, not a matter of a hard integral. Concretely, the sample mean $\bar{X}$ of Cauchy observations is itself Cauchy distributed—no better than a single observation, and it does not settle down as n grows—so even if one wrote $\bar{X} = \hat{\theta}$ the “estimator” would not converge. The contrast with Exercise 12 is worth marking: the Fisher information of the Cauchy is finite ($1/2$ per observation), so maximum likelihood estimates θ perfectly well; it is only the *moments* that fail. ■

Exercise 37

Show that the k th sample moment about the origin, based on a random sample of size n , is an unbiased estimate of the k th theoretical moment of a random variable X .

SOLUTION:

The k th sample moment about the origin is $M_k = \frac{1}{n} \sum_{i=1}^n X_i^k$. Because the X_i are identically distributed, each X_i^k has the same expectation $E\{X^k\}$ (the k th theoretical moment), assumed finite. Linearity then gives

$$E\{M_k\} = \frac{1}{n} \sum_{i=1}^n E\{X_i^k\} = \frac{1}{n} \cdot n E\{X^k\} = \boxed{E\{X^k\}}$$

so M_k is unbiased for the k th moment, for every k (whenever the moment exists). This is the principle underwriting the whole method of moments: replace each theoretical moment by its sample counterpart and one is substituting an unbiased estimate. It requires no assumption on the form of the distribution—only that the moment be finite, which is exactly where the Cauchy of Exercise 36 breaks the method. ■

Exercise 38

(a) How would you estimate $E(1/\sqrt{X})$, based on a random sample of size n , if you did not know the density of X ? (b) Compare your estimate with the maximum likelihood estimate of $E(1/\sqrt{X})$ if X possesses the density $f(x | \theta) = e^{-(x/\theta)}/\theta$, $x > 0$.

SOLUTION:

(a) The quantity is the expectation of the random variable $1/\sqrt{X}$, so estimate it by the corresponding sample average,

$$\hat{g} = \frac{1}{n} \sum_{i=1}^n \frac{1}{\sqrt{X_i}}$$

By linearity, $E\{\hat{g}\} = E\{1/\sqrt{X}\}$ whenever that expectation exists: the estimator is unbiased with no knowledge of the density at all. (This is the principle behind Exercise 37, the sample k th moment estimating the theoretical k th moment; here the function is $x^{-1/2}$ rather than x^k .)

(b) With the exponential density the target has a closed form. Substituting $u = x/\theta$, $dx = \theta du$,

$$\begin{aligned} E\left\{\frac{1}{\sqrt{X}}\right\} &= \int_0^\infty x^{-1/2} \frac{e^{-x/\theta}}{\theta} dx = \frac{1}{\theta} \int_0^\infty (\theta u)^{-1/2} e^{-u} \theta du \\ &= \theta^{-1/2} \int_0^\infty u^{-1/2} e^{-u} du = \frac{\Gamma(\frac{1}{2})}{\sqrt{\theta}} = \sqrt{\frac{\pi}{\theta}}, \end{aligned}$$

using $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. The maximum likelihood estimator of θ in this model is $\bar{X}$ (Assignment 6, Problem 3; and Exercise 18 above in the rate parameterization), and the

maximum likelihood estimate of a function of θ is that function of the estimate, so

$$\widehat{E(1/\sqrt{X})}_{\text{ML}} = \sqrt{\frac{\pi}{\bar{X}}}$$

The comparison the exercise wants: (a) is a general-purpose estimator—unbiased under any parent distribution, but it consults each observation only through $1/\sqrt{X_i}$. (b) commits to the model and channels the whole sample through the one number $\bar{X}$ that the exponential likelihood cares about; it is biased in finite samples (a nonlinear function of an average is not the average of the function), but it converges to the same target and, when the model is right, does so with smaller variance. What is bought with the model assumption is precision; what is risked is everything, since if X is not exponential the second estimator converges to $\sqrt{\pi/EX}$, which is no longer $E(1/\sqrt{X})$. ■

Exercise 39

Given $\bar{X} = 20$ for a random sample of size 25 from the density $f(x | \mu) = \frac{e^{-[(x-\mu)^2/32]}}{4\sqrt{2\pi}}$, find a 90% confidence interval for μ .

SOLUTION:

Reading the density, the exponent $-(x - \mu)^2/32 = -(x - \mu)^2/(2 \cdot 16)$ identifies the variance as $\sigma^2 = 16$, so $\sigma = 4$ is known. For a normal mean with known variance the $100(1 - \alpha)\%$ interval is $\bar{X} \pm z_{\alpha/2} \sigma/\sqrt{n}$. Here $n = 25$, $\sigma/\sqrt{n} = 4/5 = 0.8$, and for 90% the point is $z_{0.05} = 1.645$, so the half-width is $1.645 \times 0.8 = 1.316$. With $\bar{X} = 20$,

$$20 \pm 1.316, \quad \text{i.e. } (18.68, 21.32)$$

The interval is centered at the estimate and its width $2 \times 1.316 = 2.63$ is fixed by the known σ and the sample size alone—the data enter only through $\bar{X}$, shifting the interval but not its length. Exercise 40 asks what n would halve that length. ■

Exercise 40

How large a sample would be needed in Exercise 39 if the resulting confidence interval were to be half as long?

SOLUTION:

The interval's length is $2z_{\alpha/2} \sigma / \sqrt{n}$, proportional to $1/\sqrt{n}$ with everything else held fixed. To cut the length in half, $\sqrt{n}$ must double, so n must quadruple:

$$n = 4 \times 25 = 100$$

Precision in a confidence interval improves only with the square root of the sample size, so buying a factor-of-two improvement costs a factor of four in observations—the familiar diminishing return of $\sqrt{n}$ estimation. The same arithmetic governs Exercise 46's sample-size question. ■

Exercise 41

Use Table III of the appendix to obtain a random sample of size 16 of a standard normal variable. Assume that you know $\sigma = 1$ but do not know that $\mu = 0$. Use your sample values to obtain an 80% confidence interval for μ .

SOLUTION:

Reading the second column of Table III (HPS, p. 223), sixteen standard normal values are

$$\begin{aligned} &-0.329, 0.085, 0.130, -0.244, -0.882, 0.472, 0.039, 1.420, \\ &-1.033, 1.807, -0.578, 0.439, -0.852, 0.489, 0.675, -1.210, \end{aligned}$$

with $\bar{x} = 0.027$. With $\sigma = 1$ known, $n = 16$, and $z_{0.10} = 1.282$, the half-width is $1.282 \times 1/\sqrt{16} = 1.282/4 = 0.320$, so the 80% interval is

$$0.027 \pm 0.320, \quad \text{i.e. } (-0.294, 0.347)$$

The interval comfortably covers the true $\mu = 0$. Its modest confidence level (only 80%) buys the narrow width; the sample values here are reused in Exercises 43 and 44 to build intervals for σ instead. ■

Exercise 42

Given $\sum X_i = 180$ and $\sum X_i^2 = 2,000$ for a random sample of size 20 from $f(x | \sigma) = \frac{\exp[-(1/2)[(x - 10)/\sigma]^2]}{\sigma\sqrt{2\pi}}$, find a 90% confidence interval for σ^2 .

SOLUTION:

The mean is known to be $\mu = 10$, so $\sum_i (X_i - 10)^2 / \sigma^2$ is a chi-square variable with $n = 20$ degrees of freedom. First assemble the pivotal statistic from the given sums:

$$\sum_i (X_i - 10)^2 = \sum_i X_i^2 - 2(10) \sum_i X_i + 20(10)^2 = 2000 - 3600 + 2000 = 400.$$

A 90% interval brackets the χ_{20}^2 variable between its 5th and 95th percentiles, $\chi_{0.95,20}^2 = 10.851$ and $\chi_{0.05,20}^2 = 31.410$:

$$10.851 \leq \frac{400}{\sigma^2} \leq 31.410 \implies \frac{400}{31.410} \leq \sigma^2 \leq \frac{400}{10.851},$$

that is,

$$(12.73, 36.86)$$

The interval for a variance is not symmetric about a point estimate—the chi-square distribution is skewed, so the two percentiles sit at unequal distances and the interval stretches farther to the right. Knowing μ is what allows all 20 degrees of freedom; Exercise 44 pays one of them back when the mean is unknown. ■

Exercise 43

Use the sample values of Exercise 41 to obtain a 90% confidence interval for σ under the assumption that $\mu = 0$ but you do not know that $\sigma = 1$.

SOLUTION:

With $\mu = 0$ assumed known, $\sum_i X_i^2/\sigma^2 \sim \chi_{16}^2$. For the Exercise 41 sample, $\sum_i x_i^2 = 10.95$. Bracketing the χ_{16}^2 variable between $\chi_{0.95,16}^2 = 7.962$ and $\chi_{0.05,16}^2 = 26.296$ gives a 90% interval for σ^2 ,

$$\frac{10.95}{26.296} \leq \sigma^2 \leq \frac{10.95}{7.962}, \quad \text{i.e. } (0.417, 1.376),$$

and taking square roots gives the interval for σ :

$$(0.645, 1.173)$$

The true $\sigma = 1$ lies inside. Because the mean is taken as known, the pivot keeps all 16 degrees of freedom—the sharpest the sample allows. Exercise 44 works the same data without that assumption. ■

Exercise 44

Work Exercise 43 under the assumption that μ is also unknown.

SOLUTION:

When μ is unknown it must be estimated by $\bar{x}$, and the pivot becomes $\sum_i (X_i - \bar{X})^2 / \sigma^2 \sim \chi_{n-1}^2 = \chi_{15}^2$ —one degree of freedom spent on centering the data. For the Exercise 41 sample, $\bar{x} = 0.027$ and $\sum_i (x_i - \bar{x})^2 = 10.94$. Bracketing χ_{15}^2 between $\chi_{0.95,15}^2 = 7.261$ and $\chi_{0.05,15}^2 = 24.996$,

$$\frac{10.94}{24.996} \leq \sigma^2 \leq \frac{10.94}{7.261}, \quad \text{i.e. } (0.438, 1.507),$$

so for σ ,

$$(0.662, 1.228)$$

The interval is a little wider than Exercise 43's $(0.645, 1.173)$ —the price of not knowing μ . Two effects push the same way: the numerator $\sum (x_i - \bar{x})^2 \leq \sum x_i^2$ shrinks slightly (here from 10.95 to 10.94, the sample mean being nearly zero), but the degrees of freedom drop from 16 to 15, widening the chi-square percentiles' spread. The net is a modestly less precise interval, the honest cost of estimating the center from the same data used to estimate the spread. ■

Exercise 45

Given the following random sample values of a normal variable, use them to find a 95% confidence interval for σ^2 : 6.70, 4.55, 7.12, 3.30, 2.15, 4.00, 3.38, 7.93, 8.40, 5.75, 2.91, 6.50, 7.96, 3.00, 4.93.

SOLUTION:

The mean is not given, so estimate it: with $n = 15$ the sample mean is $\bar{x} = 5.24$, and the sum of squared deviations is $\sum_i (x_i - \bar{x})^2 = 61.46$. The pivot $\sum_i (X_i - \bar{X})^2 / \sigma^2$ is $\chi_{n-1}^2 = \chi_{14}^2$. A 95% interval brackets it between $\chi_{0.975,14}^2 = 5.629$ and $\chi_{0.025,14}^2 = 26.119$:

$$\frac{61.46}{26.119} \leq \sigma^2 \leq \frac{61.46}{5.629},$$

that is,

$$(2.35, 10.92)$$

The point estimate $s^2 = 61.46/14 = 4.39$ sits well inside, but far from centered—the interval reaches almost twice as far above it as below, the chi-square skewness again. With only 15 observations a variance is pinned down loosely: the upper limit is more than four times the lower, which is why variance estimation demands larger samples than mean estimation for comparable relative precision. ■

Exercise 46

In estimating the mean of a normal variable by means of a confidence interval, how large a sample will be needed if the length of a 95% confidence interval is to be less than $\sigma/10$ if σ is assumed known?

SOLUTION:

The 95% interval has length $L = 2z_{0.025} \sigma / \sqrt{n} = 2(1.96)\sigma / \sqrt{n} = 3.92 \sigma / \sqrt{n}$. Requiring $L < \sigma/10$ cancels the σ :

$$\frac{3.92}{\sqrt{n}} < \frac{1}{10} \implies \sqrt{n} > 39.2 \implies n > 1536.6.$$

Since n must be a whole number,

$$n = 1537$$

That the known σ cancels is the useful part: the required sample size depends only on the *relative* precision demanded, not on the scale of the data. Pinning the mean to within a twentieth of a standard deviation on each side takes better than fifteen hundred observations—a vivid instance of the $\sqrt{n}$ law of Exercise 40. ■

Exercise 47

Construct a 90% confidence interval for the parameter μ of the Poisson density if a sample of size 30 yielded $\sum X_i = 240$. Use the normal approximation for $\sum X_i/30$ and replace σ by its sample estimate.

SOLUTION:

The sample mean is $\bar{X} = 240/30 = 8$, the natural estimate of μ . For a Poisson, $\text{Var}(X) = \mu$, so $\text{Var}(\bar{X}) = \mu/n$, estimated by $\bar{X}/n = 8/30$; the standard error is $\sqrt{8/30} = 0.516$. By the normal approximation the 90% interval is $\bar{X} \pm z_{0.05} \cdot \text{SE} = 8 \pm 1.645(0.516)$, a half-width of 0.849:

$$8 \pm 0.849, \quad \text{i.e. } (7.15, 8.85)$$

The interval leans on the central limit theorem: a sum of 30 Poisson counts is close enough to normal that the z -based interval is trustworthy. The Poisson's mean-equals-variance property is what lets a single number, $\bar{X}$, supply both the center and (through $\bar{X}/n$) the width. ■

Exercise 48

Construct a 95% confidence interval for the parameter p of the Bernoulli density $f(x | p) = \begin{cases} p & x = 1 \\ 1 - p & x = 0 \end{cases}$ if a random sample of size 50 yielded $\sum X_i = 15$. Use the normal approximation for $\sum X_i/50$ and replace its variance by its sample estimate.

SOLUTION:

The sample proportion is $\hat{p} = 15/50 = 0.30$. For a Bernoulli, $\text{Var}(X) = p(1 - p)$, so $\text{Var}(\hat{p}) = p(1 - p)/n$, estimated by $\hat{p}(1 - \hat{p})/n = (0.30)(0.70)/50 = 0.0042$; the standard error is $\sqrt{0.0042} = 0.0648$. The 95% interval is $\hat{p} \pm z_{0.025} \cdot \text{SE} = 0.30 \pm 1.96(0.0648)$, a half-width of 0.127:

$$0.30 \pm 0.127, \quad \text{i.e. } (0.173, 0.427)$$

This is the workhorse interval for a proportion. Exercises 49 and 50 apply the same recipe to coin and die data at different confidence levels. ■

Exercise 49

Toss a coin fifty times and record the number of heads obtained. Use this value and the result in Exercise 48 to obtain a 90% confidence interval for p , which is $1/2$ here.

SOLUTION:

A seeded run of fifty tosses (in place of a physical coin) produced 27 heads, so $\hat{p} = 27/50 = 0.54$. The standard error is $\sqrt{\hat{p}(1 - \hat{p})/50} = \sqrt{(0.54)(0.46)/50} = 0.0705$, and with $z_{0.05} = 1.645$ the 90% interval is $0.54 \pm 1.645(0.0705) = 0.54 \pm 0.116$:

$$(0.424, 0.656)$$

The interval covers the true $p = 1/2$ with room to spare. Fifty tosses locate a fair coin's probability only to within about ± 0.12 at 90% confidence—a reminder of how much data a tight bound on a proportion requires. ■

Exercise 50

Toss a die sixty times and record the number of six spots obtained. Use this value and the result in Exercise 48 to obtain an 80% confidence interval for p , which is $1/6$ here.

SOLUTION:

A seeded run of sixty die tosses gave 9 sixes, so $\hat{p} = 9/60 = 0.15$. The standard error is $\sqrt{\hat{p}(1 - \hat{p})/60} = \sqrt{(0.15)(0.85)/60} = 0.0461$, and with $z_{0.10} = 1.282$ the 80% interval is $0.15 \pm 1.282(0.0461) = 0.15 \pm 0.059$:

$$(0.091, 0.209)$$

The true $p = 1/6 \approx 0.167$ sits comfortably inside. The lower confidence level (80%) and the smaller true proportion combine to give a narrower interval than Exercise 49's, though the normal approximation is working harder here—with only $\hat{p}n = 9$ successes it is near the edge of where the approximation is comfortable. ■

Exercise 51

Construct an 80% confidence interval for the parameter θ of the density $f(x | \theta) = e^{-(x/\theta)}/\theta$, $x > 0$, if a random sample of size 20 yielded $\sum X_i = 80$. Use the normal approximation for $\sum X_i/20$.

SOLUTION:

This is the exponential with mean θ , so $\bar{X} = 80/20 = 4$ estimates θ . Here $\text{Var}(X) = \theta^2$, hence $\text{Var}(\bar{X}) = \theta^2/n$, estimated by $\bar{X}^2/n = 16/20 = 0.8$; the standard error is $\sqrt{0.8} = 0.894$. With $z_{0.10} = 1.282$ the 80% interval is $\bar{X} \pm z_{0.10} \cdot \text{SE} = 4 \pm 1.282(0.894) = 4 \pm 1.147$:

$$(2.85, 5.15)$$

For the exponential the standard deviation equals the mean, so the relative precision here is set entirely by $\sqrt{n}$: the half-width is $z_{0.10} \theta / \sqrt{n}$, about 29% of $\bar{X}$ at $n = 20$. The interval is symmetric only because the normal approximation is; the exact distribution of $\bar{X}$ (a scaled chi-square) is right-skewed, so for small n a chi-square-based interval would serve better. ■

Exercise 52

Find the Bayes estimator of the Bernoulli parameter p based on a random sample of size n if $\mathcal{L}(p, a) = (a - p)^2$ and $\pi(p) = 3p(1 - p)$, $0 \leq p \leq 1$.

SOLUTION:

Under squared-error loss the Bayes estimator is the posterior mean (HPS, Thm. 2, p. 38; notes, Thm. 2.9.3). The likelihood of $s = \sum_i x_i$ successes in n Bernoulli trials is $p^s(1 - p)^{n-s}$, and the prior kernel is $p(1 - p)$ —the constant 3 is irrelevant to the posterior, since any factor not involving p cancels in the normalization. Matching kernels,

$$\pi(p \mid x) \propto p(1 - p) \cdot p^s(1 - p)^{n-s} = p^{(s+2)-1}(1 - p)^{(n-s+2)-1},$$

the $\text{Beta}(s + 2, n - s + 2)$ kernel. Its mean, shape over the sum of shapes, is the Bayes estimator:

$$\delta(X) = \frac{s + 2}{n + 4} = \frac{2 + \sum_i X_i}{n + 4}$$

The prior $\pi(p) \propto p(1 - p)$ is $\text{Beta}(2, 2)$, worth four pseudo-trials carrying two successes; it pulls the estimate toward $\frac{1}{2}$ and is outvoted by real data at rate $1/n$. This is the same conjugate machinery as Assignment 7's Bernoulli–Beta problem and the shaded rule of Assignment 1, now with a mildly informative symmetric prior in place of the uniform. ■

Exercise 53

Given $f(x | \theta) = 1/\theta$, $0 \leq x \leq \theta$, $\mathcal{L}(\theta, a) = (a - \theta)^2$, and $\pi(\theta) = \theta e^{-\theta}$, $\theta > 0$, (a) calculate the marginal density $f(x)$, (b) use $f(x)$ to find the posterior density $f(\theta | x)$, (c) use the result obtained in (b) to calculate $E(\theta | x)$.

SOLUTION:

The single observation X is uniform on $[0, \theta]$, and the prior on θ is $\text{Gamma}(2, 1)$. The joint density is where the arithmetic turns friendly:

$$f(x, \theta) = f(x | \theta)\pi(\theta) = \frac{1}{\theta} \cdot \theta e^{-\theta} = e^{-\theta}, \quad 0 \leq x \leq \theta, \theta > 0,$$

the factor $1/\theta$ from the likelihood cancelling the θ in the prior.

(a) The marginal integrates out θ , which must exceed x for the observation to be possible:

$$f(x) = \int_x^\infty e^{-\theta} d\theta = e^{-x}, \quad x > 0,$$

so X is marginally standard exponential.

(b) The posterior is the ratio,

$$f(\theta | x) = \frac{f(x, \theta)}{f(x)} = \frac{e^{-\theta}}{e^{-x}} = e^{-(\theta-x)}, \quad \theta > x$$

a unit exponential shifted to start at x : given the data, θ is x plus an $\text{Exp}(1)$ amount.

(c) Its mean is that shift plus the exponential's mean of 1; formally, with $u = \theta - x$,

$$E(\theta | x) = \int_x^\infty \theta e^{-(\theta-x)} d\theta = \int_0^\infty (u+x)e^{-u} du = 1+x,$$

so the Bayes estimator is

$$\delta(X) = E(\theta \mid X) = X + 1$$

The observation gives a hard lower bound— θ cannot be below the value already seen—and the prior supplies the rest, adding exactly one expected unit above it. A clean case where a non-conjugate-looking setup collapses to a shifted exponential because the likelihood and prior were built to cancel. ■

Exercise 54

Find the Bayes estimator of the Poisson parameter μ based on a random sample of size n if $\mathcal{L}(\mu, a) = (a - \mu)^2$ and $\pi(\mu) = e^{-\mu}$, $\mu > 0$.

SOLUTION:

Under squared-error loss the Bayes estimator is the posterior mean (HPS, Thm. 2, p. 38; notes, Thm. 2.9.3). Each observation has pmf $e^{-\mu}\mu^{x_i}/x_i!$, so the likelihood of the sample is

$$L(\mu) = \prod_{i=1}^n \frac{e^{-\mu}\mu^{x_i}}{x_i!} = \frac{e^{-n\mu}\mu^s}{\prod_i x_i!},$$

and the factorials do not involve μ . Matching kernels,

$$\begin{aligned}\pi(\mu \mid x) &\propto e^{-\mu} \cdot e^{-n\mu}\mu^s && \text{(prior times likelihood)} \\ &= \mu^{(1+s)-1} e^{-(n+1)\mu}, && \text{(add exponents)}\end{aligned}$$

the Gamma($1 + s$, $n + 1$) kernel in the shape–rate convention. Its mean is the shape over the rate, so

$$\delta(X) = \frac{1 + \sum_i X_i}{n + 1}$$

This is Assignment 7's Poisson–gamma estimator $(\alpha + s)/(\beta + n)$ at $\alpha = \beta = 1$, the prior $e^{-\mu}$ being exactly Gamma($1, 1$): worth one prior observation carrying one event, and outvoted by the data at rate $1/n$. ■

Exercise 55

Find the Bayes estimator of the parameter θ for the density $f(x | \theta) = \theta e^{-\theta x}$, $x > 0$, based on a single observed value of X , if

$$\mathcal{L}(\theta, a) = (a - \theta)^2 \quad \text{and} \quad \pi(\theta) = 1, \quad 0 < \theta < 1.$$

SOLUTION:

The posterior is proportional to prior times likelihood on the prior's support:

$$\pi(\theta | x) \propto \theta e^{-\theta x}, \quad 0 < \theta < 1,$$

a $\text{Gamma}(2, x)$ kernel cut off at 1—not a standard family, so this time the normalizer and the mean must be computed. Both integrals yield to integration by parts. For the normalizer, with $u = \theta$ and $dv = e^{-\theta x} d\theta$,

$$\int_0^1 \theta e^{-\theta x} d\theta = \left[-\frac{\theta e^{-\theta x}}{x} \right]_0^1 + \frac{1}{x} \int_0^1 e^{-\theta x} d\theta = -\frac{e^{-x}}{x} + \frac{1 - e^{-x}}{x^2} = \frac{1 - (1 + x)e^{-x}}{x^2},$$

and for the numerator, with $u = \theta^2$,

$$\begin{aligned} \int_0^1 \theta^2 e^{-\theta x} d\theta &= \left[-\frac{\theta^2 e^{-\theta x}}{x} \right]_0^1 + \frac{2}{x} \int_0^1 \theta e^{-\theta x} d\theta \\ &= -\frac{e^{-x}}{x} + \frac{2[1 - (1 + x)e^{-x}]}{x^3} = \frac{2 - (x^2 + 2x + 2)e^{-x}}{x^3}, \end{aligned}$$

the last step putting both terms over x^3 : $-x^2 e^{-x} + 2 - 2(1 + x)e^{-x} = 2 - (x^2 + 2x + 2)e^{-x}$.

The Bayes estimator is the ratio,

$$\delta(X) = \mathbb{E}\{\theta \mid X\} = \frac{2 - (X^2 + 2X + 2)e^{-X}}{X [1 - (1 + X)e^{-X}]}$$

Two checks confirm the formula behaves. As $x \rightarrow 0$ the posterior tends to $\pi(\theta \mid x) \propto \theta$ on $(0, 1)$, whose mean is $\int_0^1 \theta^2 d\theta / \int_0^1 \theta d\theta = \frac{1/3}{1/2} = \frac{2}{3}$, and the boxed expression has that limit; as $x \rightarrow \infty$ the truncation stops binding and $\delta(x) \approx 2/x$, the mean of the untruncated $\text{Gamma}(2, x)$. A large observation argues for a small rate, and the estimator follows it down—but never past the prior’s floor at 0. ■

Exercise 56

Given $f(x | \theta) = e^{-(x/\theta)}/\theta$, $x > 0$, $\pi(\theta) = (2/3)\theta$, $1 \leq \theta \leq 2$, and $\mathcal{L}(\theta, a) = (a - \theta)^2$, (a) calculate $\mathcal{R}(\theta, d)$ for $d_1(X) = X$ and $d_2(X) = X - 1$, and determine which is the minimax estimator with respect to these two estimators, (b) calculate the Bayes solution with respect to these two estimators.

SOLUTION:

The single observation X is exponential with mean θ , so $E\{X\} = \theta$ and $\text{Var}(X) = \theta^2$.

Risk is mean squared error, $\mathcal{R}(\theta, d) = E(d - \theta)^2 = \text{Var}(d) + [\text{bias}]^2$.

(a) For $d_1 = X$: unbiased, so $\mathcal{R}(\theta, d_1) = \text{Var}(X) = \theta^2$. For $d_2 = X - 1$: bias $E\{X - 1\} - \theta = -1$ and the same variance θ^2 , so $\mathcal{R}(\theta, d_2) = \theta^2 + 1$. Thus d_2 's risk exceeds d_1 's by 1 at every θ — d_1 dominates. The minimax choice compares worst-case risks on $[1, 2]$:

$$\max_{1 \leq \theta \leq 2} \mathcal{R}(\theta, d_1) = 2^2 = 4, \quad \max_{1 \leq \theta \leq 2} \mathcal{R}(\theta, d_2) = 2^2 + 1 = 5,$$

so

$$d_1(X) = X \text{ is minimax among the two}$$

(b) The Bayes solution minimizes the Bayes risk $r(\pi, d) = \int_1^2 \mathcal{R}(\theta, d) \pi(\theta) d\theta$. For d_1 ,

$$r(\pi, d_1) = \int_1^2 \theta^2 \cdot \frac{2}{3} \theta d\theta = \frac{2}{3} \left[\frac{\theta^4}{4} \right]_1^2 = \frac{2}{3} \cdot \frac{15}{4} = \frac{5}{2},$$

and for d_2 , since its risk is larger by 1,

$$r(\pi, d_2) = r(\pi, d_1) + \int_1^2 1 \cdot \frac{2}{3} \theta d\theta = \frac{5}{2} + \frac{2}{3} \cdot \frac{3}{2} = \frac{5}{2} + 1 = \frac{7}{2}.$$

The smaller Bayes risk again belongs to d_1 :

$$d_1(X) = X \text{ is the Bayes solution among the two, } r(\pi, d_1) = \frac{5}{2}$$

Both criteria point the same way here, and for the plainest of reasons: d_2 subtracts a constant from an already unbiased estimator, inflating its risk uniformly, so no averaging or worst-casing could rescue it. The exercise's value is procedural—it shows the minimax and Bayes calculations side by side on a restricted class where the answer is unambiguous. ■

Exercise 57

Given $f(x \mid \theta) = 1/\theta$, $0 \leq x \leq \theta$, $\pi(\theta) = 1$, $0 < \theta < 1$, and $\mathcal{L}(\theta, a) = |a - \theta|$, (a) calculate $\mathcal{R}(\theta, d)$ and $r(\pi, d)$ for $d(X) = X$; (b) calculate $\mathcal{R}(\theta, d)$ and $r(\pi, d)$ for $d(X) = X^2$ and compare with the results in (a).

SOLUTION:

The loss is absolute error, and X is uniform on $[0, \theta]$ with θ uniform on $(0, 1)$.

(a) For $d(X) = X$: since $X \leq \theta$ always, $|X - \theta| = \theta - X$, so

$$\mathcal{R}(\theta, X) = E|X - \theta| = \int_0^\theta (\theta - x) \frac{dx}{\theta} = \frac{1}{\theta} \left[\theta x - \frac{x^2}{2} \right]_0^\theta = \frac{\theta}{2}.$$

The Bayes risk averages this over the prior:

$$r(\pi, X) = \int_0^1 \frac{\theta}{2} d\theta = \boxed{\frac{1}{4}}.$$

(b) For $d(X) = X^2$: because $0 < \theta < 1$, we have $X^2 \leq \theta^2 < \theta$, so again the loss loses its absolute value, $|X^2 - \theta| = \theta - X^2$, and with $E\{X^2\} = \int_0^\theta x^2 dx / \theta = \theta^2/3$,

$$\mathcal{R}(\theta, X^2) = \theta - E\{X^2\} = \theta - \frac{\theta^2}{3}.$$

The Bayes risk is

$$r(\pi, X^2) = \int_0^1 \left(\theta - \frac{\theta^2}{3} \right) d\theta = \frac{1}{2} - \frac{1}{9} = \boxed{\frac{7}{18}} \approx 0.389.$$

Comparison: $r(\pi, X) = \frac{1}{4} = 0.25 < \frac{7}{18} = 0.389 = r(\pi, X^2)$, so $d(X) = X$ is the better rule under absolute-error loss. In fact the domination is pointwise— $\mathcal{R}(\theta, X) -$

$\mathcal{R}(\theta, X) = \frac{\theta}{2} - \frac{\theta^2}{3} = \theta(\frac{1}{2} - \frac{\theta}{3}) > 0$ throughout $(0, 1)$ —so squaring the observation only hurts. Both estimators sit *below* θ (a single uniform draw and its square both underestimate the endpoint), and X^2 , being smaller, sits farther below; the natural fix, as Exercise 2 showed, is to scale X up rather than power it down. ■

Exercise 58

Given $f(x | \theta) = \frac{x^{\alpha-1} e^{-\theta x} \theta^\alpha}{\Gamma(\alpha)}$, $x > 0$, squared error loss, and $\pi(\theta) = \frac{\theta^{a-1} e^{-\beta \theta} \beta^a}{\Gamma(a)}$, where α , a , and β are given positive constants, calculate the Bayes estimator of θ based on a single observed value of X .

SOLUTION:

As a function of θ , the likelihood is $f(x | \theta) \propto \theta^\alpha e^{-\theta x}$ (the $x^{\alpha-1}/\Gamma(\alpha)$ is a constant in θ), and the prior is the $\text{Gamma}(a, \beta)$ kernel $\theta^{a-1} e^{-\beta \theta}$. Under squared-error loss the Bayes estimator is the posterior mean. Multiplying and adding exponents,

$$\pi(\theta | x) \propto \theta^\alpha e^{-\theta x} \cdot \theta^{a-1} e^{-\beta \theta} = \theta^{(\alpha+a)-1} e^{-(\beta+x)\theta},$$

the $\text{Gamma}(\alpha + a, \beta + x)$ kernel in the shape–rate convention. Its mean is shape over rate, so

$$\delta(X) = E(\theta | X) = \frac{\alpha + a}{\beta + X}$$

The Gamma prior is conjugate to this Gamma likelihood in its rate parameter: the single observation adds α to the shape and X to the rate, exactly as one Poisson count would in Assignment 7's Poisson–gamma model. The prior contributes a pseudo-shape and β pseudo-rate; as those shrink toward zero the estimator tends to α/X , which is the maximum likelihood estimator of θ from this density (since $E\{X\} = \alpha/\theta$). ■

Exercise 59

Given $f(x, y) = 8xy$, $0 \leq x \leq 1$, $0 \leq y \leq x$, (a) find the equation of the best predictor $d(x) = E(Y | x)$; (b) what can be said about the best linear predictor for this problem?

SOLUTION:

(a) The conditional expectation requires the conditional density, which requires the marginal. Integrating the joint over the range of y at fixed x ,

$$f_X(x) = \int_0^x 8xy \, dy = 8x \cdot \frac{x^2}{2} = 4x^3, \quad 0 \leq x \leq 1,$$

so

$$f(y | x) = \frac{f(x, y)}{f_X(x)} = \frac{8xy}{4x^3} = \frac{2y}{x^2}, \quad 0 \leq y \leq x,$$

which integrates to $\int_0^x (2y/x^2) \, dy = x^2/x^2 = 1$, as a conditional density must. Then

$$d(x) = E(Y | x) = \int_0^x y \cdot \frac{2y}{x^2} \, dy = \frac{2}{x^2} \cdot \frac{x^3}{3} = \boxed{\frac{2x}{3}}$$

(b) The best predictor came out *linear* in x , so the best linear predictor of the book's Formula (31) cannot do better than to coincide with it—and it does. One can confirm this directly from the moments. Using $f_X(x) = 4x^3$: $E\{X\} = \int_0^1 x \cdot 4x^3 \, dx = \frac{4}{5}$, $E\{X^2\} = \frac{2}{3}$, so $\text{Var}(X) = \frac{2}{3} - \frac{16}{25} = \frac{2}{75}$; and since $E(Y | X) = 2X/3$, $E\{Y\} = \frac{2}{3}E\{X\} = \frac{8}{15}$ and $E\{XY\} = E\{X \cdot \frac{2}{3}X\} = \frac{2}{3}E\{X^2\} = \frac{4}{9}$, giving $\text{Cov}(X, Y) = \frac{4}{9} - \frac{4}{5} \cdot \frac{8}{15} = \frac{4}{225}$. The best linear predictor's slope and intercept are then

$$\text{slope} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{4/225}{2/75} = \frac{2}{3}, \quad \text{intercept} = E\{Y\} - \frac{2}{3}E\{X\} = \frac{8}{15} - \frac{2}{3} \cdot \frac{4}{5} = 0,$$

so the best linear predictor is $\frac{2}{3}x$ —*exactly* the best predictor. Hence:

the best linear predictor equals the best predictor; the regression is exactly linear

There is no approximation error and nothing lost by restricting to lines. This is the special situation §2.9 of the text describes, and it happens here by an accident of the density: the support $0 \leq y \leq x$ scales with x and the conditional density $2y/x^2$ has the same triangular shape on every slice, so its mean sits at the same fraction, two-thirds of the way up. Exercise 60 is the instructive opposite, where $E(Y \mid x)$ is a genuine curve and the line only approximates it. ■

Exercise 60

Given $f(x, y) = \frac{2x + 4y}{3}$, $0 \leq x \leq 1$, $0 \leq y \leq 1$, (a) find the equation of the best predictor $E(Y | x)$; (b) find the equation of the best linear predictor given by (31); (c) graph these two predictors to see how well the linear predictor approximates the best predictor.

SOLUTION:

(a) The marginal of X is

$$f_X(x) = \int_0^1 \frac{2x + 4y}{3} dy = \frac{2x + 2}{3} = \frac{2(x + 1)}{3}, \quad 0 \leq x \leq 1,$$

so the conditional density is $f(y | x) = (2x + 4y)/(2(x + 1)) = (x + 2y)/(x + 1)$, and

$$E(Y | x) = \int_0^1 y \cdot \frac{x + 2y}{x + 1} dy = \frac{1}{x + 1} \left[\frac{x}{2} + \frac{2}{3} \right] = \boxed{\frac{3x + 4}{6(x + 1)}}$$

a genuine rational function of x , not a line.

(b) The best linear predictor needs the first two moments of each variable and their covariance. From $f_X(x) = 2(x + 1)/3$: $E\{X\} = \frac{5}{9}$, $E\{X^2\} = \frac{7}{18}$, so $\text{Var}(X) = \frac{7}{18} - \frac{25}{81} = \frac{13}{162}$. From $f_Y(y) = (1 + 4y)/3$: $E\{Y\} = \frac{11}{18}$. And $E\{XY\} = \iint xy \frac{2x + 4y}{3} dx dy = \frac{1}{3}$, so $\text{Cov}(X, Y) = \frac{1}{3} - \frac{5}{9} \cdot \frac{11}{18} = -\frac{1}{162}$. The slope and intercept of Formula (31) are

$$\text{slope} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{-1/162}{13/162} = -\frac{1}{13},$$

$$\text{intercept} = E\{Y\} - \text{slope} \cdot E\{X\} = \frac{11}{18} + \frac{1}{13} \cdot \frac{5}{9} = \frac{17}{26},$$

so the best linear predictor is

$$\boxed{\ell(x) = \frac{17}{26} - \frac{1}{13}x}$$

Both predictors slope gently *downward*: the negative covariance means larger x is associated with slightly smaller Y on average.

(c) The two curves are drawn below on $0 \leq x \leq 1$. They agree closely—the best predictor is only mildly nonlinear—parting most near the ends of the range.

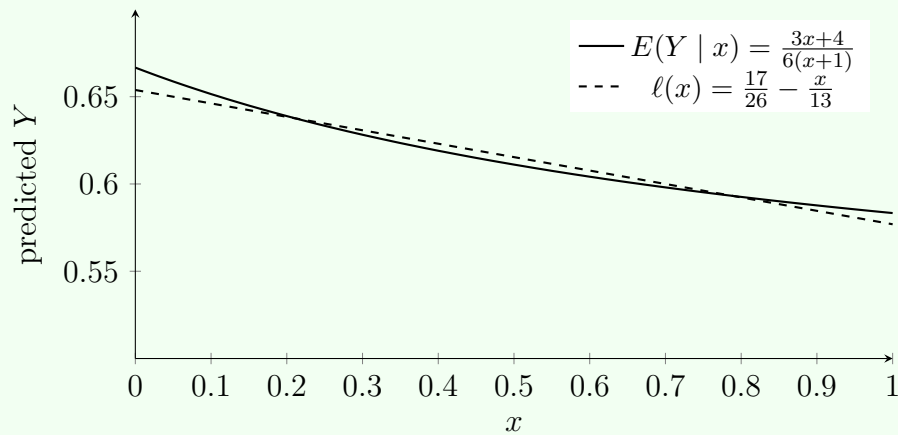


Figure 2. BEST PREDICTOR VERSUS BEST LINEAR PREDICTOR. The exact conditional mean (solid) and its linear approximation from Formula (31) (dashed) on $[0, 1]$. The two run within about 0.01 of each other everywhere; the line is an excellent stand-in here even though, unlike Exercise 59, the regression is not exactly linear.

At the endpoints the gap is small: $E(Y | 0) = \frac{2}{3} \approx 0.667$ against $\ell(0) = \frac{17}{26} \approx 0.654$, and $E(Y | 1) = \frac{7}{12} \approx 0.583$ against $\ell(1) = \frac{15}{26} \approx 0.577$. The best linear predictor minimizes mean squared error *among lines*; that it tracks the true curve this well is a feature of this joint density's mild dependence, and is exactly the comparison Exercise 59 could not display because there the two predictors were identical. ■

Exercise 61

Suppose you performed an experiment m times and obtained s successes. Use this information to find the posterior density of p given s if the prior density of p is uniform. If the experiment is performed n additional times and S successes are obtained, what will the Bayes estimate become using squared error if the posterior density of p is treated as the prior density for the second set of experiments? How does this compare with combining the two experiments and using the uniform prior density?

SOLUTION:

First experiment. The uniform prior is $\text{Beta}(1, 1)$, with density 1 on $(0, 1)$, and the likelihood of s successes in m Bernoulli trials is $p^s(1-p)^{m-s}$ (the product $\prod_i p^{x_i}(1-p)^{1-x_i}$ with exponents summed). Matching kernels,

$$\pi(p \mid s) \propto 1 \cdot p^s(1-p)^{m-s} = p^{(1+s)-1}(1-p)^{(1+m-s)-1},$$

so the posterior is

$$p \mid s \sim \text{Beta}(1 + s, 1 + m - s)$$

Second experiment, with this posterior as the new prior. The new likelihood is $p^S(1-p)^{n-S}$, and the exponents add once more:

$$\pi(p \mid s, S) \propto p^{(1+s)-1}(1-p)^{(1+m-s)-1} \cdot p^S(1-p)^{n-S} = p^{(1+s+S)-1}(1-p)^{(1+m-s+n-S)-1},$$

the $\text{Beta}(1 + s + S, 1 + m + n - s - S)$ kernel. Under squared-error loss the Bayes estimate is its mean,

$$\delta = \frac{1 + s + S}{2 + m + n}$$

Combining instead. Pooling the data into one experiment of $m + n$ trials with $s + S$ successes and using the uniform prior gives, by the first computation with (m, s) replaced by $(m + n, s + S)$, the posterior $\text{Beta}(1 + s + S, 1 + (m + n) - (s + S))$ —the same distribution, hence the same estimate. The comparison the exercise asks for:

sequential and batch updating give the same posterior and the same estimate

Nothing about this is special to the Beta family: the sequential posterior is proportional to $\text{prior} \times \text{likelihood}_1 \times \text{likelihood}_2$, and multiplication does not care how the factors are grouped. The posterior after the first experiment is a complete record of what those m trials had to say, which is why yesterday's posterior can serve as today's prior without loss. It is a fitting last exercise for the chapter: the shaded estimator $(1 + s)/(2 + m)$ that entered this course in Assignment 1 as an unexplained formula leaves it as the running total of a Bayesian ledger that can be settled in any order. ■

References

- Blitzstein, Joseph K., and Jessica Hwang. *Introduction to Probability*. 2nd ed., CRC Press, Boca Raton, 2019.
- DeGroot, Morris H., and Mark J. Schervish. *Probability and Statistics*. 4th ed., Addison-Wesley, Boston, 2012.
- Hoel, Paul G., Sidney C. Port, and Charles J. Stone. *Introduction to Probability Theory*. Houghton Mifflin, Boston, 1971. (“Volume I.”)
- Hoel, Paul G., Sidney C. Port, and Charles J. Stone. *Introduction to Statistical Theory*. Houghton Mifflin, Boston, 1971. (Cited as HPS.)
- Pitman, Jim. *Probability*. Springer-Verlag, New York, 1993.
- Rabinowitz, Daniel. Module 2 lecture slides and *Assignment 9*, STAT S4204: Statistical Inference. Columbia University, Summer 2026.
- Rabinowitz, Daniel, and Tyler Sotomayor. *Notes in Statistical Inference: Lecture Notes for STAT S4204*. Columbia University, Summer 2026. <https://tylersotomayor.github.io/stat4204/>. (Cited as notes.)
- Ross, Sheldon M. *A First Course in Probability*. 9th ed., Pearson, Boston, 2014.